

Cyber Security

VULNERABILITY ASSESSMENT REPORT (TASK 1)

Website: <http://httpforever.com>

Date: April 26, 2026

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For: Future Interns



EXECUTIVE SUMMARY:

I tested <http://httpforever.com> for common security weaknesses. The website has 5 problems that need attention.

● **THE BOTTOM LINE**

The website has two separate security issues with how it handles visitors:

- The HTTP version works but has no encryption. Anyone on the same WiFi can see what you're doing on the site. The browser marks it as "Not Secure."
- The HTTPS version has an expired certificate. When visitors try to use the secure version, their browser shows "Your connection is not private" and blocks access completely. Most people will leave immediately.

● **WHAT NEEDS TO BE FIXED**

Two critical issues must be fixed within 24 hours:

- Replace the expired SSL certificate so HTTPS works properly
- Force all visitors to use HTTPS automatically (redirect from HTTP)

Three other issues are less urgent but still important:

- Remote access (SSH) is open to everyone on the internet
 - An email port (SMTP) is open and could be abused
 - The web server software is outdated
-

Introduction:

This report presents the results of a vulnerability assessment performed on <http://httpforever.com> on April 26, 2026.

The following tools were used:

- Nmap for port and service detection
- SecurityHeaders.com for security header analysis
- Browser DevTools for manual verification

For each finding, this report explains:

- What the issue is
 - Why it matters to your business
 - The risk level (High/Medium/Low)
 - How to fix it
-

Vulnerabilities:

FINDINGS TABLE				
Findings	Risk Level	Issues	Impact	Fix
No HTTPS Redirect	High Risk	Website serves HTTP without redirecting to HTTPS	Data can be intercepted on public WiFi	Configure 301 redirect from HTTP to HTTPS
Invalid SSL Certificate	High Risk	HTTPS certificate is invalid (ERR_CERT_DATE_INVALID)	Visitors see security warnings; no encryption protection	Install valid SSL certificate from Let's Encrypt
SSH Service Exposed	Medium Risk	Port 22 (SSH) open to entire internet	Attackers can attempt to brute force server access	Restrict SSH to specific IP addresses
SMTP Service Exposed	Medium Risk	Port 25 (email service) is open	Could be abused to send spam	Block port 25 at firewall
Outdated Software	Low Risk	nginx 1.18.0 and OpenSSH 8.9p1 detected	Known vulnerabilities may exist for these versions	Update to latest stable versions

Evidence (Screenshots):

1, notsecure.png (HTTP Warning)



Figure 1: HTTP Version Security Warning

The website loads over unencrypted HTTP. The browser displays "Not Secure" in the address bar, indicating that any data transmitted can be intercepted by attackers on the same network.

Risk: HIGH

2, httpsversionpng.png (HTTPS Certificate Error)

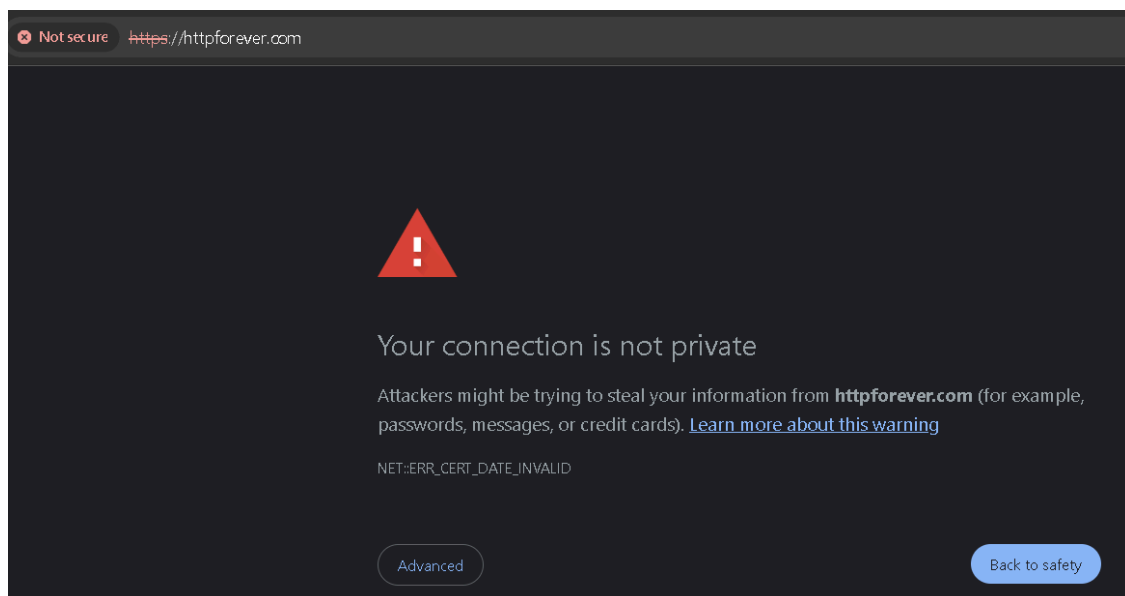


Figure 2: HTTPS Certificate Error

EER_CERT_DATE_INVALID. This indicates the SSL certificate is expired or invalid.

Risk: HIGH

3, nmap-version-screenshot.png (Version Scan)

```
C:\Windows\System32>nmap -sV httpforever.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-26 01:05 +0400
Nmap scan report for httpforever.com (146.190.62.39)
Host is up (0.27s latency).
Other addresses for httpforever.com (not scanned): 2604:a880:4:1d0::1f1:2000
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 8.9p1 Ubuntu 3ubuntu0.13 (Ubuntu Linux; protocol 2.0)
25/tcp    open  smtp?
80/tcp    open  http         nginx 1.18.0 (Ubuntu)
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
443/tcp   open  ssl/http     nginx 1.18.0 (Ubuntu)
1 service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.cgi?new-service :
SF-Port25-TCP:V=7.99%I=7%O=4/26%Time=69ED2C93%P=1686-pc-windows-windows%r(
SF:NULL,22,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(Help,22
SF:,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(GenericLines,2
SF:2,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(GetRequest,22
SF:,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(HTTPOptions,22
SF:,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(RTSPRequest,22
SF:,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(RPCCheck,22,"5
SF:54\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(DNSVersionBindReq
SF:TCP,22,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(Terminal
SF:ServerCookie,22,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r
SF:(TLSSessionReq,22,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")
SF:%r(Kerberos,22,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(
SF:SMBProgNeg,22,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(L
SF:ANDES-RC,22,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(NC
SF:P,22,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(JavaRMI,22
SF:,"554\x205\.3\.0\x20connection\x20not\x20allowed\r\n")%r(giop,22,"554\x
SF:205\.3\.0\x20connection\x20not\x20allowed\r\n");
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.25 seconds
```

Figure 3: Nmap Service Version Detection

The scan reveals server software versions:

- nginx 1.18.0 (web server)
- OpenSSH 8.9p1 (SSH service)

Both versions are outdated and may have known vulnerabilities.

Risk: MEDIUM (exposed services) + LOW (outdated versions)

4, nmap-screenshot.png (Basic Port Scan)

```
C:\Windows\System32>nmap httpforever.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-26 00:59 +0400
Nmap scan report for httpforever.com (146.190.62.39)
Host is up (0.27s latency).
Other addresses for httpforever.com (not scanned): 2604:a880:4:1d0::1f1:2000
Not shown: 994 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
25/tcp    open  smtp
80/tcp    open  http
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 6.16 seconds
```

Figure 4: Nmap Port Scan Results

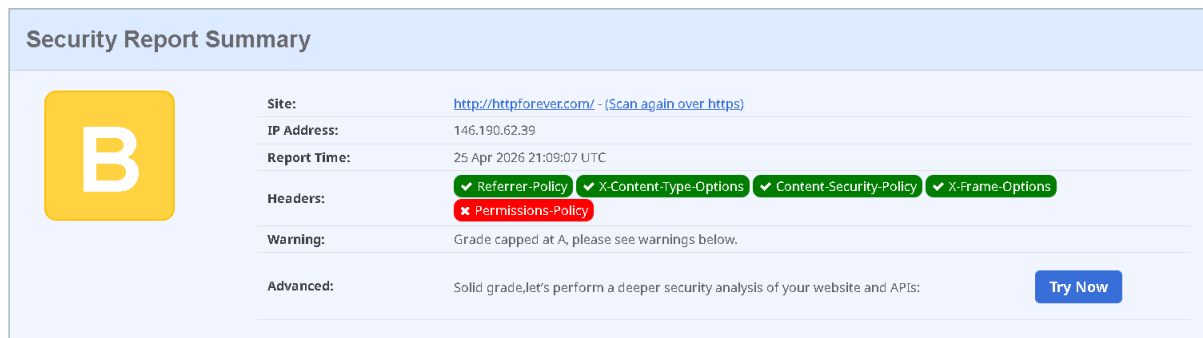
The scan shows open ports:

- Port 22 (SSH) - Remote administration
- Port 25 (SMTP) - Email service
- Port 80 (HTTP) - Unencrypted web traffic
- Port 443 (HTTPS) - Encrypted web traffic (with invalid certificate)

Multiple exposed services increase the attack surface.

Risk: MEDIUM

5, security-headers.png (Headers Scan)



The image shows a 'Security Report Summary' for the website <http://httpforever.com/>. The report includes the following details:

- Site:** <http://httpforever.com/> - (Scan again over https)
- IP Address:** 146.190.62.39
- Report Time:** 25 Apr 2026 21:09:07 UTC
- Headers:**
 - ✓ Referrer-Policy
 - ✓ X-Content-Type-Options
 - ✓ Content-Security-Policy
 - ✓ X-Frame-Options
 - ✗ Permissions-Policy
- Warning:** Grade capped at A, please see warnings below.
- Advanced:** Solid grade, let's perform a deeper security analysis of your website and APIs. [Try Now](#)

Figure 5: Security Headers Analysis

The security headers scan shows:

- Referrer-Policy (present)
- X-Content-Type-Options (missing)
- Content-Security-Policy (present)
- X-Frame-Options (present)

Only one header is missing, which is better than most websites.

Risk: LOW

Remediation:

Priority 1 - HIGHEST (Fix within 24 hours)

- **Remediation:**

" Redirect all HTTP traffic to HTTPS."

- **Explanation:**

" This ensures every visitor automatically uses the secure version of your website. No one can accidentally use the unencrypted version. It's a one-time setting change on your web server."

Finding 2: Invalid SSL Certificate (HIGH RISK)

- **Remediation:**

" Install a valid, unexpired SSL certificate."

- **Explanation:**

" An SSL certificate is like a digital ID for your website. When it's expired, browsers block your site and show scary warnings. Replacing it with a valid certificate removes these warnings and enables secure connections."

Finding 3: SSH Service Exposed (MEDIUM RISK)

- **Remediation:**

" Restrict SSH access to only trusted IP addresses."

- **Explanation:**

" SSH is the tool used to manage your server from anywhere. Leaving it open to the whole internet means attackers can try to guess your password from anywhere in the world. Limiting it to your office IP address blocks everyone else."